

Read the article "The Challenge of Exploring Venus." Then answer the questions.

The Challenge of Exploring Venus

- 1 Venus, sometimes called the "Evening Star," is one of the brightest points of light in the night sky, making it simple for even an amateur stargazer to spot. However, this nickname is misleading since Venus is actually a planet. In our solar system, Venus is the second planet from our sun. While Venus is simple to see from the distant but safe vantage point of Earth, it has proved a very challenging place to examine more closely.
- 2 Often referred to as Earth's "twin," Venus is the closest planet to Earth in terms of density and size, and occasionally the closest in distance too. Earth, Venus, and Mars, our other planetary neighbor, orbit the sun at different speeds. These differences in speed mean that sometimes we are closer to Mars and other times to Venus. Because Venus is sometimes right around the corner—in space terms—humans have sent numerous spacecraft to land on this cloud-draped world. Each previous mission was unmanned, and for good reason, since no spacecraft survived the landing for more than a few hours. Maybe this issue explains why not a single spaceship has touched down on Venus in more than three decades. Numerous factors contribute to Venus's reputation as a challenging planet for humans to study, despite its proximity to us.
- 3 A thick atmosphere of almost 97 percent carbon dioxide blankets Venus. Even more challenging are the clouds of highly corrosive sulfuric acid in Venus's atmosphere. On the planet's surface, temperatures average over 800 degrees Fahrenheit, and the atmospheric pressure is 90 times greater than what we experience on our own planet. These conditions are far more extreme than anything humans encounter on Earth; such an environment would crush even a submarine accustomed to diving to the deepest parts of our oceans and would liquefy many metals. Also notable, Venus has the hottest surface temperature of any planet in our solar system, even though Mercury is closer to our sun. Beyond high pressure and heat, Venusian geology and weather present additional impediments like erupting volcanoes, powerful earthquakes, and frequent lightning strikes to probes seeking to land on its surface.
- 4 If our sister planet is so inhospitable, why are scientists even discussing further visits to its surface? Astronomers are fascinated by Venus because it may well once have been the most Earth-like planet in our solar system. Long ago, Venus was probably covered largely with oceans and could have supported various forms of life, just like Earth. Today, Venus still has some features that are analogous to those on Earth. The planet has a surface of rocky sediment and includes familiar features such as valleys, mountains, and craters. Furthermore, recall that Venus can sometimes be our nearest option for a planetary visit, a crucial consideration given the long time frames of space travel. The value of returning to Venus seems indisputable, but what are the options for making such a mission both safe and scientifically productive?
- 5 The National Aeronautics and Space Administration (NASA) has one particularly compelling idea for sending humans to study Venus. NASA's possible solution to the hostile conditions on the surface of Venus would allow scientists to float above the fray. Imagine a blimp-like vehicle hovering 30 or so miles above the rolling Venusian landscape. Just as our jet airplanes travel at a higher altitude to fly over many storms, a vehicle hovering over Venus would avoid the unfriendly ground conditions by staying up and out of their way. At thirty-plus miles above the surface, temperatures would still be toasty at around 170 degrees Fahrenheit, but the air pressure would be close to that of sea level on Earth. Solar power would be plentiful, and radiation would not exceed Earth levels. Not easy conditions, but survivable for humans.
- 6 However, peering at Venus from a ship orbiting or hovering safely far above the planet can provide only limited insight on ground conditions because most forms of light cannot penetrate the dense atmosphere, rendering standard forms of photography and videography ineffective. More importantly, researchers cannot take samples of rock, gas, or anything else, from a distance. Therefore, scientists seeking to conduct a thorough mission to understand Venus would need to get up close and personal despite the risks. Or maybe we should think of them as challenges. Many researchers are working on innovations that would allow our machines to last long enough to contribute meaningfully to our knowledge of Venus.
- 7 NASA is working on other approaches to studying Venus. For example, some simplified electronics made of silicon carbide have been tested in a chamber simulating the chaos of Venus's surface and have lasted for three weeks in such conditions. Another project is looking back to an old technology called mechanical computers. These devices were first envisioned in the 1800s and played an important role in the 1940s during World War II. The thought of computers existing in those days may sound shocking, but these devices make calculations by using gears and levers and do not require electronics at all. Modern computers are enormously powerful, flexible, and quick, but tend to be more delicate when it comes to extreme physical conditions. Just imagine exposing a cell phone or tablet to acid or heat capable of melting tin. By comparison, systems that use mechanical parts can be made more resistant to pressure, heat, and other forces.
- 8 Striving to meet the challenge presented by Venus has value, not only because of the insight to be gained on the planet itself, but also because human curiosity will likely lead us into many equally intimidating endeavors. Our travels on Earth and beyond should not be limited by dangers and doubts but should be expanded to meet the very edges of imagination and innovation.

Prewriting/Planning

Use the blank Prewriting/Planning space below for notes, lists, webs, outlines, or anything else that might help you plan your writing. If you need additional space for planning, raise your hand and your teacher will give you a sheet of scratch paper. **You must write your final draft in the box on the right.**



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Editing Checklist

1. Check your capitalization and punctuation.
2. Spell all words correctly.
3. Check for sentence fragments or run-on sentences.
4. Keep verb tense consistent.
5. Make sure subject and verb agree.
6. Use words according to the rules of Standard English.
7. Remember to paragraph correctly.

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In “The Challenge of Exploring Venus,” the author suggests that studying Venus is a worthy pursuit despite the dangers it presents. Using details from the article, write an essay evaluating how well the author supports this idea.

Be sure to include

- a claim that evaluates how well the author supports the idea that studying Venus is a worthy pursuit despite the dangers
- an explanation of the evidence from the article that supports your claim
- an introduction, a body, and a conclusion to your essay

Final Draft

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